

INTERNSHIPS	Aerospace Engineering Research Intern <i>Princeton University</i>	2025 – Present
	Designed and implemented a custom diffusion model to predict thrust-control vectors for global trajectory search in the Circular Restricted Three-Body Problem (CR3BP).	
	Astrophysics Research Intern <i>Harvard-Smithsonian CfA</i>	2024 – Present
	Developing new methods for precision tracking of interstellar objects and NEOs using JWST and terrestrial telescopes.	
	Solid Propulsion Team Member <i>MIT Rocket Team</i>	2024 – Present
	Manufactured rocket nozzles and conducted hydrostatic and static firing tests for the Spaceport America Cup competition.	
	Engineer & Researcher <i>Institute for Earth Observations</i>	2021 – Present
	- Developed ground stations and designed lowcost CubeSats for international distribution; showcased at IEEE, SmallSat Conference, and AGU. - Collaborated with NASA, U.S. Naval Academy, Kenyan Space Agency, and more; impacted STEM curricula globally.	
	Machine Learning Research Intern <i>Harvard-Smithsonian CfA</i>	2024 – 2024
	Designed a multimodal (infrared video & audio) transformer architecture for UAP analysis, integrating AST and MViT with DINO pretraining and InfoSieve.	
	Machine Learning Research Intern <i>National University of Singapore</i>	2022 – 2024
	Conducted NLP, computational linguistics, and AI bias research on transformer models; presented at international conferences.	
	Machine Learning Engineering Intern <i>Fireflies.ai</i>	2021 – 2023
	Integrated GPT-3 and advanced transformer models for automated meeting summarization, robust Q&A systems, and additional NLP tasks.	
	Astrophysics Researcher <i>Summer Science Program (New Mexico Tech)</i>	2022 – 2022
	Computed Keplerian elements of asteroid 2001 MZ7 using the Method of Gauss and Monte Carlo simulations; published by IAU MPC.	
	NASA SEES Intern <i>NASA & UT Austin</i>	2021 – 2021
	Trained and evaluated YOLOv4 CNNs for precise mosquito habitat detection; first author at AGU, Globe IVSS, arXiv, and NASA symposium.	
	Astrobiology Research Intern <i>ASSIP (George Mason University)</i>	2021 – 2022
	Simulated ETI probabilities via Monte Carlo; lead author at NASA's AbSciCon22 on extraterrestrial intelligence modeling.	
	WSSP Researcher <i>Rutgers University</i>	2021 – 2021
	Analyzed <i>Landoltia punctata</i> DNA sequences; published two mRNA sequences in NCBI GenBank and achieved course valedictorian.	

EDUCATION	Harvard College	Cambridge, MA
	<i>S.B. in Mechanical and Aerospace Engineering, Astrophysics</i>	2023 – 2027
	• Activities: Harvard Libertarian Association (President & Founder), I.C.D.P. Fellowship, Franklin Fellowship, Emerging Technology Group (NASA Artemis), Harvard Innovation Labs, Student Astronomers at Harvard-Radcliffe	
	Massachusetts Institute of Technology	Cambridge, MA
	<i>Cross-Enrollment</i>	2025 – 2027
	• Activities: MIT Rocket Team (Solid Propulsion)	
	Cinnaminson High School	Cinnaminson, NJ
	<i>High School Diploma</i>	2019 – 2023
	• Academics: SAT: 1570, Rank: 4/207, GPA: 105.3782 • Activities: Class Vice President, Founder & President (Science and Engineering Research Club), Design Lead (Robotics), Founder & President (TSA), National Honor Society, World Affairs Council, Global Economic Forum	
AWARDS AND HONORS	• HCRP Research Fellow, Harvard College	2024, 2025
	• Congressional Award Gold Medal, U.S Congress	2023
	• Congressional App Challenge: 1st Place, U.S. House of Representatives	2021
	• Carson Scholar, Carson Scholars Fund	2022
	• South Jersey Achievement, South Jersey Magazine	2022
	• Hall of Fame, Cinnaminson High School	2023
	• AP Scholar with Distinction, College Board	2022
	• National Merit Commended Student, College Board	2022
	• HOBY Youth Leadership Delegate, Cinnaminson High School	2021
	• Voice of Democracy (School & District): 1st Place, VFW	2021

- PUBLICATIONS
1. **Elango, S.**, Cuffe, S., Palma, A., Acchione, Q., Wu, D. Genetic Algorithm Identification of Initial Orbits for Interplanetary and Earth–Moon Transfers. *AIAA Conference Paper (AIAA 2026-2923), Trajectory/Mission/Maneuver Design and Optimization III*, 2026.
 2. **Elango, S.**, Loeb, A. Astrometric Parallax Measurements with JWST for Localization of Near-Earth Objects. *arXiv preprint*, 2025.
 3. Moore, J. D., Kramer, M., Gallant, J., Khatri, A., **Elango, S.** Creating the Earth SySTEM Observatory: Integrating Spaceborne and Ground-Based Environmental Research. *IEEE ISEC, Princeton University*, 2025.
 4. **Dominé, L.**, Elango, S., Kricheli, J., Fedorenko, A., Brenard, T., Jacoby, D. Multi-Modal Generalized Class Discovery for Scalable Autonomous All-Sky Surveys. *IAIFI Summer Workshop, Massachusetts Institute of Technology*, 2024.
 5. Moore, J., **Elango, S.**, Friedman, M., Kang, J., Maceo, C. The A3Sat Emulator: A Catalyst in Disruptive CubeSat and Space Technology. *IEEE ISEC, Princeton University*, 2024.
 6. **Elango, S.**, Moore, J., Friedman, M. A3Sat: Expanding STEM Education through CubeSats and a Global Data Network. *AGU Annual Meeting*, 2023.
 7. Moore, J., Friedman, M., **Elango, S.**, Kang, J., Maceo, C. A3Sat Emulator Enterprise: To Observe the Earth and Visualize the Future. *Small Satellite Conference*, 2023.
 8. **Elango, S.**, Jaidka, K. Data-Driven Unsupervised Semantic Network Algorithm for Large Language Model Examination/Exploration. *SSRN*, 2023.
 9. **Elango, S.**, Jaidka, K. The GPT-Comparator: Discovering and Reporting Spatial and Topical Biases in Generative Pre-Trained Transformers. *SSRN*, 2023.
 10. Moore, J., Friedman, M., **Elango, S.**, Kang, J., Maceo, C. A3Sat: Using CubeSat Emulators to Broaden Advanced Participation in STEM Education. *IEEE ISEC, Johns Hopkins University (APL)*, 2023.
 11. **Elango, S.**, Feng, E., Melendrez, E. Observations of Near-Earth Asteroid 2001 MZ7. *IAU Minor Planet Center*, 2022.
 12. **Elango, S.**, Jaidka, K. A Multi-Modal Architecture for Identifying Anti-Social Imagery. *ICWSM*, 2022.
 13. **Elango, S.**, Summers, M. Estimation of Extraterrestrial Intelligent Civilizations and Attributes per Exoplanetary Continuum through Algorithmic Simulation and Civilization Modeling. *AGU AbSciCon*, 2022.
 14. Moore, J., **Elango, S.**, Friedman, M., Kang, J. A3Sat: Using CubeSats to Inspire the Next Generation STEM Professionals. *IEEE ISEC, Princeton University*, 2022.
 15. **Elango, S.**, Ramachandran, N. Novel Approach to Autonomous Mosquito Habitat Detection Using Satellite Imagery and Convolutional Neural Networks for Disease Risk Mapping. *AGU Fall Meeting*, 2021.
 16. **Elango, S.** Landoltia punctata clone W377.20 urease accessory protein-like mRNA. *NCBI GenBank*, 2021.
 17. **Elango, S.** Landoltia punctata clone W378.20 ribulose-1,5-bisphosphate carboxylase small subunit-like mRNA. *NCBI GenBank*, 2021.